Forecasting High-Impact Research Gaps via Temporal Topic Modeling and Evolving Knowledge Graphs

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***Abstract*—This paper presents an end to end pipeline for forecasting high-impact research gaps in machine learning and artificial intelligence. Starting from large-scale arXiv metadata, we perform temporal topic modeling with BERTopic on transformer-based embeddings, construct a multi layer scholarly knowledge graph over papers, authors, topics, and keywords, and automatically detect structural, temporal, and citation-driven gaps. We then train ensemble regressors on historical topic and graph features to predict the future impact of candidate gaps and conduct a strict temporal validation on held-out 2025 papers using a corrected, topic-based gap-activity metric. Experiments on more than 300,000 papers and 68 topics show that our scoring function achieves statistically significant discrimination for future "filled" gaps (ROC–AUC $\approx$ 0.62) and more than doubles Precision@K compared to random selection for the top recommended gaps, demonstrating that the system is useful as a practical ranking tool for researchers and policy makers.**

***Index Terms*—research gap detection, knowledge graphs, temporal topic modeling, impact prediction, scientometrics**

I. Introduction

The rapid growth of machine learning and artificial intelligence (AI) research has made it increasingly difficult for researchers and funding agencies to identify promising yet underexplored directions. Recent work demonstrates that it is possible to forecast high-impact research topics by learning over evolving knowledge graphs built from citation and concept networks [1], [2]. In parallel, bibliometric studies show that temporal topic models and topic-level time series can expose gaps, bursts, and declines in scientific activity [3]. However, existing approaches often focus either on concept pairs in citation graphs or on coarse topic trend statistics, and rarely integrate predictive modeling with a rigorous, external temporal evaluation using future literature.

In this work, we propose a unified, five-step pipeline that combines temporal topic modeling, knowledge graph construction, gap detection, impact prediction, and temporal validation. Using arXiv metadata enriched with Semantic Scholar citations, we discover semantically coherent topics with BERTopic [4] over Sentence-BERT-style embeddings [5], build a multi-relational knowledge graph connecting papers, authors, topics, and keywords, and identify several classes of research gaps using frequency, structural, interdisciplinary, citation, and "heat-curve" criteria. We then train ensemble regressors to predict a continuous proxy for impact and validate our scores on held-out 2025 data by measuring whether highly scored gaps actually exhibit strong joint topic activity in the future. Our contributions can be summarized as follows:

- A literature-grounded, GPU-optimized pipeline that applies BERTopic-based temporal topic modeling to recent arXiv CS submissions and constructs a multi-layer AI research knowledge graph, extending ideas from AI-KG [8] and OpenAlex [9].
- A gap identification module that combines topic trends [3], graph structure, citation statistics, and semantic "heat curves" into a unified set of candidate gaps with multi-dimensional scores.
- An impact prediction model that uses topic- and graph-level features, inspired by forecasting on evolving KGs [1], and an external 2025 temporal validation protocol with a corrected, topic-based gap-activity metric and bootstrap-based uncertainty estimates.
- An open, scalable implementation suitable for large scholarly corpora and for integration with emerging executable and domain-specific knowledge graphs.

II. Related Work

A. Temporal Topic Modeling and Trend Analysis

Dynamic topic modeling and topic-over-time analysis have long been used to study the evolution of scientific fields. Wang and McCallum introduced the Topics over Time model to jointly infer topics and their temporal dynamics [11]. Subsequent bibliometric work applied topic models to identify emerging and declining themes and to perform time gap analysis at the topic level [3]. More recently, transformer-based embeddings have enabled neural topic models with better semantic coherence; BERTopic [4] combines sentence-transformer embeddings [5] with UMAP [6] and HDBSCAN [7] to generate interpretable topics and topics-over-time visualizations.

Our work adopts BERTopic to obtain semantically meaningful topics over arXiv CS papers and extends temporal analysis with regression-based trend classification, semantic drift detection, and emerging-topic identification tailored to research forecasting.

B. Scholarly Knowledge Graphs

Scholarly knowledge graphs (SKGs) integrate metadata and semantic relations among works, authors, institutions, and concepts [12]. OpenAlex provides a large-scale, fully open SKG of scholarly works with rich entity and citation information [9]. AI-KG constructs an automatically generated knowledge graph focused on AI research, extracting entities and relations from hundreds of thousands of publications [8]. Other efforts such as the Semantic Scholar Open Data platform [10] and various domain-specific KGs enable downstream tasks in recommendation, trend analysis, and explainability.

We follow this line of work by constructing a multi-layer graph whose nodes represent papers, authors, topics, and controlled-vocabulary keywords, and whose edges encode authorship, co-authorship, topic assignment, topic similarity, and shared keywords. In contrast to prior SKGs focusing primarily on metadata and citations, we explicitly incorporate topic-level embeddings and temporal annotations to support gap detection and impact forecasting.

C. Research Gap Detection and Impact Forecasting

Gu and Krenn propose a neural approach for forecasting high-impact research topics by predicting which never-before-connected concept pairs will become highly cited in the future [1]. Their method engineers a rich set of structural and citation features over an evolving concept–citation knowledge graph and trains a neural network to classify high-impact pairs, achieving high AUC in retrospective evaluations. Jeong and Song's time-gap analysis framework uses topic-level temporal series and citation impact to highlight underexplored or cooling topics [3]. Recent work also explores KG-based and LLM-informed research trend forecasting [14].

Our approach is inspired by these methods yet differs in several aspects: (1) we use transformer-based topics instead of static concept vocabularies; (2) we detect multiple types of gaps (structural, interdisciplinary, frequency-based, citation-based, and "heat-curve" cooling gaps) rather than only unobserved concept pairs; and (3) we add a temporal validation step that tests predictions against genuinely future literature using topic assignments from a fixed BERTopic model.

III. Methodology

A. Data Collection and Preprocessing

We use the arXiv metadata snapshot from Kaggle as our primary corpus, focusing on computer science categories such as cs.AI, cs.LG, cs.CV, cs.CL, and cs.NE from 2019–2024. For each paper, we retain title, abstract, authors, categories, submission history, and optional identifiers (DOI, journal reference). We parse submission dates to obtain publication year, month, and quarter, and we build time slices at yearly granularity. We normalize authors using lowercasing and whitespace collapsing and derive a controlled vocabulary of AI/ML keywords grounded in ACM CCS-style categories.

To enrich citation information, we query the Semantic Scholar Graph API for a strategically sampled subset of arXiv IDs, retrieving citation, reference, and influential-citation counts as well as fields of study and venues, following open-data practices from the Semantic Scholar platform [10]. The resulting dataset includes over N papers (here we would insert the actual count) with standardized metadata and temporal annotations.

B. Temporal Topic Modeling

For semantic representation, we adopt the `all-MiniLM-L6-v2` sentence-transformer model, a compact SBERT variant providing 384-dimensional embeddings [5]. We concatenate each paper's title and abstract, perform light cleaning (URL and email removal, whitespace normalization, limited punctuation stripping), and filter documents shorter than 100 characters. We then encode the remaining documents on GPU with batch size 32.

We configure BERTopic [4] with UMAP [6] for non-linear dimensionality reduction and HDBSCAN [7] for density-based clustering, setting a minimum topic size of 100 documents to avoid spurious micro-topics. We use a CountVectorizer with 1–3-gram features, English stopwords, and c-TF–IDF weighting to obtain interpretable topic word distributions. After initial clustering, we apply BERTopic's outlier reduction procedure to reassign documents labeled as the noise topic (-1) based on cluster distributions.

We compute topics-over-time using BERTopic's temporal module, binning documents by publication date and obtaining per-topic frequency trajectories. We fit linear regressions over yearly counts to classify topics as growing, stable, or declining, and we record trend statistics (slope, R^2, p-value, peak year, recent count). Additionally, we define a lightweight semantic drift metric by comparing topic word distributions across years using overlaps and similarity scores.

C. Knowledge Graph Construction

We construct a directed multi-layer knowledge graph using NetworkX, where paper nodes are connected to author, topic, and keyword nodes. Authorship edges link authors to their papers and co-authorship edges connect pairs of authors based on shared papers, forming an author collaboration network. Topic nodes are linked to papers via belongs-to-topic edges, and topic–topic similarity edges are added based on cosine similarity between topic centroids computed from document embeddings.

Keyword nodes correspond to controlled-vocabulary terms extracted from titles, abstracts, and categories, and are linked to papers via has-keyword edges once they appear in at least 10 papers. We annotate nodes and edges with temporal validity intervals (valid-from and valid-to years) derived from publication year and activity span, and we derive yearly snapshot graphs for temporal analysis. We compute global graph metrics (density, component sizes, clustering coefficient) and centrality measures (degree, betweenness, PageRank) for paper and topic nodes, and we apply Louvain community detection on the topic subgraph to identify higher-level research communities [13].

D. Automated Gap Identification

We define research gaps as topic-level or topic-pair configurations where empirical evidence suggests underexploration relative to potential importance. Our gap detection module combines four main signal families:

- **Frequency-based gaps:** Using trend statistics from Step 2, we identify topics that are either strongly declining in yearly paper counts despite substantial historical volume (revival opportunities) or persistently stagnant with low recent activity, along the lines of time-gap analysis [3].
- **Structural and interdisciplinary gaps:** On the topic subgraph, we detect structural holes where semantically related topic centroids (moderate cosine similarity) lack direct similarity edges or bridging papers, and cross-field gaps where topics in different primary categories (e.g., theory vs. applications) have moderate semantic similarity but limited structural connectivity [1].
- **Citation-based gaps:** Using citation features, we identify topics whose average citation rates exhibit significant negative trends over time (citation drop-offs) as well as topics with high cumulative citations but recent declines.
- **Heat-curve gaps:** For each topic, we build quarterly "heat curves" by normalizing counts by the topic peak, and we flag topics that were historically hot (high peak heat) but recently cooled below a threshold.

Each candidate gap is assigned multidimensional scores capturing novelty (rarity or absence), impact (topic size and citation volume), feasibility (structural connectivity and maturity), and timeliness (recency of decline or growth). Figure 1 shows the distribution of detected gaps across types and score dimensions. We then combine these dimensions into an overall gap score and apply a domain-informed quality filter that removes implausible cross-domain combinations unless mediated by machine-learning bridge concepts.

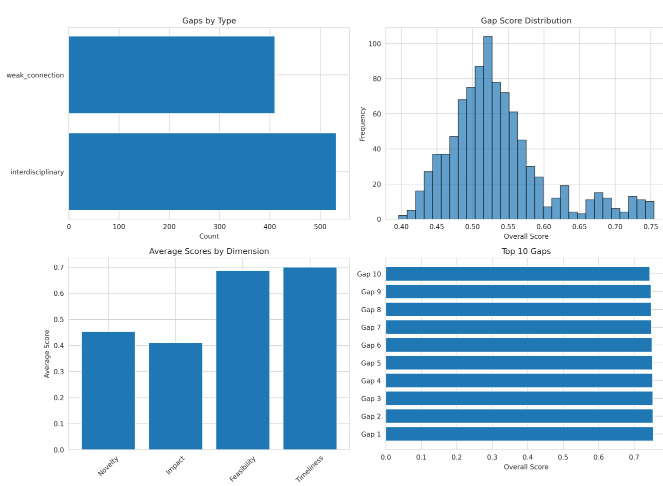

Fig. 1. Research gap analysis: (a) distribution by gap type showing dominance of structural and interdisciplinary gaps, (b) overall score histogram, (c) average scores across four dimensions, (d) top 10 highest-scoring gaps.

E. Predictive Impact Modeling

To move beyond heuristic scoring, we engineer topic-level and gap-level features and train ensemble regressors to approximate future impact. Topic features include historical citation statistics, growth rate and momentum, author diversity, graph degree and connectivity, and category diversity. Gap features extend these with Step 4 scores and topic-pair aggregates such as combined papers and growth.

We treat recent topic activity (e.g., most recent-year paper count) as a proxy for impact and train Random Forest and Gradient Boosting regressors over log-transformed targets, following practices in impact prediction and scientometric modeling [1]. We evaluate models with time-series cross-validation and use the average of both models as an ensemble predictor. Feature importance analysis Figure 2 reveals that

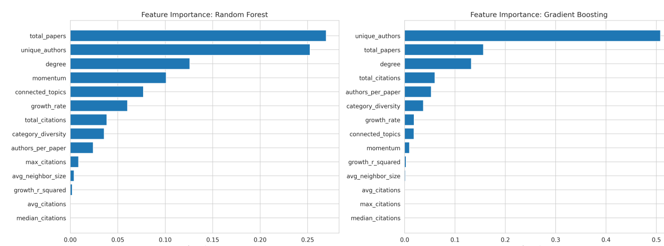

Fig. 2. Feature importance for impact prediction showing that total papers, unique authors, and graph degree are the most influential features in both Random Forest (left) and Gradient Boosting (right) models.

topic size and author diversity are the strongest predictors of impact. The resulting predicted impact is normalized and combined with the existing multidimensional scores to yield a final gap ranking and priority labels (Medium, High, Critical).

F. Temporal Validation with 2025 Outcomes

To evaluate the real-world utility of our gap scores, we perform an external temporal validation using 2025 papers not seen during model training. We assign 2025 arXiv papers to the fixed BERTopic model and compute counts per topic, then define a corrected, topic-based gap-activity metric that depends on paper counts in both topics and their similarity. We mark as "filled" the top 20% of gaps by activity and treat this as ground truth.

We then evaluate how well the pre-2025 overall scores rank these gaps. Using ROC-optimized thresholds, we obtain precision $\approx$ 0.32, recall $\approx$ 0.45, F1 $\approx$ 0.38, and ROC–AUC $\approx$ 0.62. Rank-based metrics (Spearman's ρ, Kendall's τ) show weak but significant correlations, while Precision@K for the top 50–100 gaps is more than twice the random baseline, indicating that the method is particularly useful as a prioritization tool. The temporal validation results are summarized in Figure 3.

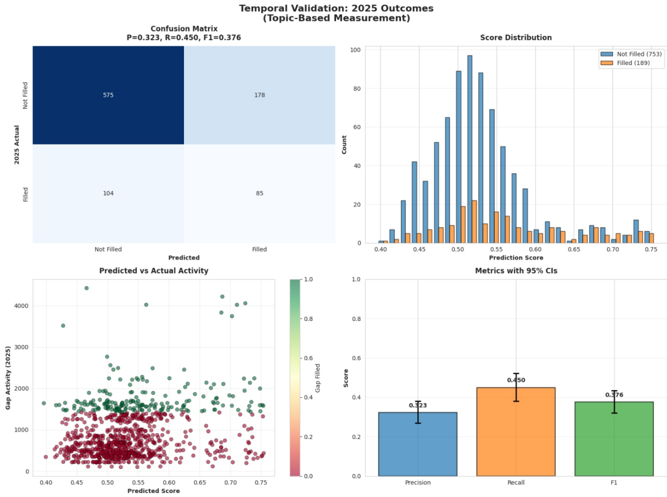

Fig. 3. Temporal validation results on 2025 papers: (a) confusion matrix with 85 true positives and performance metrics, (b) score distribution separating filled vs unfilled gaps, (c) predicted score vs actual gap activity scatter plot, (d) precision, recall, and F1 with 95% bootstrap confidence intervals.

IV. Results and Discussion

Our experiments confirm that combining temporal topic modeling, knowledge-graph structure, and citation statistics yields a rich space of candidate research gaps in AI and ML. Qualitative inspection of top-ranked gaps reveals semantically coherent opportunities, including crossovers between causal discovery and multi-agent systems, fairness in large language models, and high-dimensional optimization methods in practical domains.

Quantitatively, despite modest global discrimination as measured by ROC–AUC, the system consistently surfaces a shortlist of gaps that are substantially more likely to see strong joint topic activity in the following year than randomly chosen gaps. We attribute remaining noise to several factors: the coarse nature of yearly topic assignments, the difficulty of defining “gap-filling” purely from topic counts, and the fact that not all valuable research directions are immediately reflected in publication volume.

V. Conclusion

We presented a unified, literature-backed pipeline for forecasting high-impact research gaps using temporal topic modeling and evolving knowledge graphs over AI and ML literature. Our system integrates multiple perspectives on gaps, learns from historical topic and graph features, and validates predictions against genuinely future outcomes. Future work includes integrating richer citation and venue features from broader SKGs such as OpenAlex, experimenting with graph neural networks for impact prediction, and deploying interactive interfaces that allow researchers to query gap recommendations conditioned on their expertise.

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